

torchgeo-bench

Two-Week Sprint Recap

~50 PRs · GPU evaluation · OlmoEarth v1.1 · calibration · efficiency profiling · leaderboards

June 2026

What shipped

GPU GPU KNN via faissknn

FAISS-backed KNN with optional CUDA path. Auto-fallback to CPU.

CLI Typer + Rich

Replaced argparse + tqdm. Beautiful progress bars, rich tracebacks.

PROFILE Efficiency Profiling

Throughput, GFLOPs, peak GPU mem, \$/inference. Pareto front: accuracy vs cost.

MODELS OlmoEarth v1 + v1.1

nano→large, plus v1.1 linear-embed family. DINOv3-SAT ViT-L web-pretrained.

CALIB Calibration Metrics

ECE · RMS-CE · MCE + temperature scaling on every probe.

DATA EuroSAT Spatial Split

Geographically disjoint train/test. Harder than random split.

QUALITY Cleanlab Label Audit

Label-quality scores across all GeoBench V1+V2 datasets.

POOL CLS + Mean Pool Ablations

TerraTorch/ScaleMAE/Clay — pool=cls|mean|both sweep.

FIX Silent-bug Sweep

Removed try/except covers; fixed minmax_zscore, fp16 overflow, label gaps.

SEG Patch-Linear Head

ViT-native segmentation decoder. Segmentation probing now first-class.

MULTI SAR + Landsat Modalities

OlmoEarth mixed-sensor support, auto input-resolution.

GPU KNN — faissknn

PR #53

PR #55

PR #89

PR #94

PR #101

CPU path:

```
1  clf = KNNClassifier(n_neighbors=5, device="cpu")
2  clf.fit(x_train, y_train)
3  preds = clf.predict(x_test)
```

GPU path (opt-in via `pip install -e ".[cuda]"`):

```
1  clf = KNNClassifier(
2      n_neighbors=5,
3      device="cuda:0",
4      metric="cosine", # l2 | ip | cosine
5  )
6  clf.fit(x_train, y_train)
7  preds = clf.predict(x_test)
8  proba = clf.predict_proba(x_test)
```

Key fixes:

- `n_classes = max(y)+1` **not** `len(unique(y))` — avoids `IndexError` on partitions with missing class labels
- `use_fp16=False` in evaluation — raw sensor DN values (~10 000) overflow fp16 L2 distances → random KNN
- `faiss-cpu` is now the **default**; `faiss-cuda-cu128` moved to `[cuda]` extra — fixes install on macOS + non-manylinux Linux (#120)

Config knobs (#94 , #99):

```
1  eval:
2      knn_k: 5          # neighbours
3      knn_device: null  # null → inherit cfg.device
4                        # "cpu" forces faiss-cpu KNN
```

Typer + Rich CLI

PR #88

Replaced `argparse` + `tqdm` with **Typer** + **Rich** across the whole CLI.

Typed commands

`torchgeo-bench run`, `torchgeo-bench download` — auto-generated `--help`, type-checked args, Hydra passthrough for config overrides.

Rich progress bars

Live extraction progress with ETA and throughput instead of plain `tqdm` spinners.

Rich tables + tracebacks

Dataloader tuning results rendered as formatted tables. Full color tracebacks with local variable context on errors.

Efficiency Profiling

PR #60

PR #62

PR #63

PR #67

Metrics recorded per model run:

```
1  {
2    # GPU
3    "throughput_samples_per_sec": 1420.3,
4    "peak_gpu_mem_gb":          3.2,
5    "gflops":                    61.6,
6    "params_m":                  307.4,
7    # CPU
8    "throughput_samples_per_sec_cpu": 42.1,
9    # Cost
10   "cost_usd_per_1M_samples":    0.12,
11 }
```

GFLOPs via FlopCounterMode:

```
1  with torch.profiler.FlopCounterMode() as fc:
2      model(sample)
3      gflops = fc.get_total_flops() / 1e9
```

Throughput:

```
1  # warm-up, then timed forward passes
2  sps = n_samples / elapsed_s
```

Cost extrapolation:

```
1  # A100 spot ~$1.50/hr on Lambda
2  cost = (1_000_000 / throughput) / 3600 * 1.50
```

Explorer shows Pareto front: accuracy vs throughput / cost.

OlmoEarth Integration

[PR #84](#)[PR #85](#)[PR #93](#)

Config (nano → large):

```
1 # conf/model/olmoearth_v1_base.yaml
2 _target_: torchgeo_bench.models.OlmoEarthBench
3 name: olmoearth_v1_base
4 variant: base
5 normalization: identity # model handles its own
```

Auto-rescale to S2 DN:

```
1 class OlmoEarthBench(BenchModel):
2     expected_input_unit = InputUnit.S2_DN
3
4     def _forward_patch_features(
5         self, images: Tensor, **_
6     ) → Tensor:
7         # model_native normalizer already
8         # rescaled inputs to S2 DN range
9         return self.backbone(images)
```

m-eurosat Linear Accuracy:

	Model	Score
1	OlmoEarth Large	0.976
2	OlmoEarth Base	0.975
3	DOFA Large	0.973
4	OlmoEarth v1.1 Base	0.970
5	DINOv3-SAT ViT-L	0.969

Dominates S2 datasets — EuroSAT, BigEarthNet, So2Sat.
Nano/Tiny competitive at a fraction of the size.

OlmoEarth v1.1

PR #99

What changed (v1 → v1.1):

- Linear patch embedding (vs. convolutional)
- Single bandset per modality
- Updated masking + loss functions

≈ 3× FEWER MACS at comparable accuracy.

```
1 # conf/model/olmoearth_v1_1_base.yaml
2 name: olmoearth_v1_1_base
3 version: v1.1 # selects weight family
```

30/30 sweep tasks · 0 failures · Nano/Tiny/Base.

Linear probe — v1 vs v1.1

Dataset	Metric	v1	v1.1
m-eurosat (Base)	Acc	.975	.970
m-so2sat (Base)	Acc	.720	.728
m-so2sat (Tiny)	Acc	.656	.693
m-bigearthnet (Tiny)	mAP	.691	.717
benv2 (Tiny)	mAP	.817	.826
treesatai (Base)	mAP	.647	.645

Smaller variants gain most — Tiny/Nano jump on So2Sat + BigEarthNet (KNN So2Sat Tiny .506 → **.567**). Slight EuroSAT dip. Net: same accuracy at a third of the compute.

CLS Token + Pool Ablations

PR #73

PR #74

PR #75

PR #76

Added `pool=cls|mean|both` sweep across TerraTorch/ScaleMAE/Clay wrappers.

Finding: token choice matters dataset-to-dataset.

- **CLS** wins on ForestNet, Brick Kiln — tasks with strong global structure
- **Patch-mean** wins on EuroSAT, BigEarthNet — spatially distributed labels
- **Terramind** has no CLS token — `_cls` configs dropped (#76)

m-brick-kiln Linear Accuracy:

	Model	Score
1	DINOv3-SAT ViT-L	0.976
2	Clay v1.5 Base+CLS	0.975
3	DOFA Base	0.974
4	DOFA Large	0.974
5	Clay v1.5 Base	0.972

Patch-Linear Head

PR #124

New segmentation decoder wired into

`SegmentationProbe` :

```
1 probe = SegmentationProbe(  
2     model=backbone,  
3     head_type="patch_linear", # new  
4 )
```

`PatchLinearHead` — lightweight ViT decoder. Projects each patch token directly to pixel logits:

```
1 ChannelLayerNorm  
2 → Conv2d(D, C × P², 1)  
3 → pixel_shuffle(P)  
4 → bilinear resize (if needed)
```

No skip connections, no upsampling pyramid — purely linear.

Why it matters:

SIMPLE

No skip connections, no upsampling pyramid. Purely linear — interpretable and fast.

VIT-NATIVE

Designed for ViTs where spatial tokens are the primary representation. Complements the existing `LinearHead` (classification) decoder.

FLEXIBLE

Handles arbitrary output sizes via bilinear resize fallback — works across datasets regardless of patch/image size mismatch.

Previously only `head_type="linear"` (classification) was available. Segmentation probing is now a first-class evaluation path.

EuroSAT Spatial Split + Cleanlab

PR #50

PR #52

Geographically disjoint train/test:

```
1 # Standard: random split
2 # → models exploit spatial autocorrelation
3
4 # Spatial: tiles from different regions
5 # → true generalization test
6 ds = EuroSATSpatialBench(
7     root=DATA_ROOT,
8     partition="default",
9     bands=["B02", "B03", "B04", "B08"],
10 )
```

OlmoEarth Large KNN: **0.959** vs 0.956 random split

OlmoEarth Base Linear: **0.978** vs 0.975 random split

Cleanlab label audit:

```
1 from cleanlab.filter import find_label_issues
2
3 issues = find_label_issues(
4     labels=y_train,
5     pred_probs=pred_proba,
6     return_indices_ranked_by=
7         "self_confidence",
8 )
9 # → flags likely mislabeled samples
```

Applied across all GeoBench V1+V2. Results in

[results/cleanlab/](#) .

Surfaces annotation noise in BigEarthNet, ForestNet, TreeSatAI — useful for reweighting or curriculum training.

Leaderboard — EuroSAT

RESULTS

m-eurosat (Accuracy)

	Model	KNN	Lin
1	OlmoEarth Large	.956	.976
2	OlmoEarth Base	.946	.975
3	Panopticon	.948	.968
4	DOFA Large	.936	.973
5	OlmoEarth v1.1 Base	.930	.970

eurosat-spatial (Accuracy)

	Model	KNN	Lin
1	OlmoEarth Large	.959	.977
2	OlmoEarth Base	.941	.978
3	Panopticon	.930	.962
4	OlmoEarth Tiny	.928	.963
5	ResNet50-RGB MoCo	.933	.958

Leaderboard — BigEarthNet

RESULTS

m-bigearthnet (micro-mAP)

	Model	KNN	Lin
1	OlmoEarth v1.1 Base	.664	.771
2	OlmoEarth Large	.664	.764
3	OlmoEarth Base	.658	.769
4	Panopticon	.652	.735
5	Terramind Large	.628	.750

benv2 (micro-mAP)

	Model	KNN	Lin
1	OlmoEarth Base	.735	.853
2	OlmoEarth v1.1 Base	.734	.852
3	OlmoEarth Large	.728	.850
4	Terramind Large	.712	.846
5	OlmoEarth v1.1 Tiny	.725	.826

Leaderboard — ForestNet + So2Sat

RESULTS

m-forestnet (Accuracy)

	Model	KNN	Lin
1	DINOv3-SAT ViT-L	.425	.582
2	Panopticon	.427	.550
3	ScaleMAE Large+CLS	.403	.569
4	Clay v1.5 Base+CLS	.414	.556
5	Clay v1.5 Base	.395	.551

Hard dataset — all models below .60

m-so2sat (Accuracy)

	Model	KNN	Lin
1	OlmoEarth v1.1 Base	.606	.728
2	OlmoEarth Base	.577	.720
3	OlmoEarth v1.1 Tiny	.567	.693
4	OlmoEarth Large	.562	.689
5	Panopticon	.532	.693

Linear leader: Terramind Base .739 — biggest KNN/Lin gap in the set

Leaderboard — Brick Kiln + PV4GER + TreeSatAI

RESULTS

m-brick-kiln (Acc)

	Model	KNN	Lin
1	DOFA Large	.969	.974
2	DOFA Base	.965	.974
3	Clay Base	.964	.972
4	Clay Base+CLS	.960	.975
5	OlmoEarth Base	.945	.970

m-pv4ger (Acc)

	Model	KNN	Lin
1	DINOv3 ViT-L	.964	.974
2	DOFA Large	.967	.969
3	DOFA Base	.965	.966
4	ScaleMAE+CLS	.956	.970
5	Panopticon	.958	.968

treesatai (mAP)

	Model	KNN	Lin
1	DINOv3-SAT ViT-L	.477	.682
2	Clay Base+CLS	.475	.671
3	DINOv3 ViT-L	.473	.664
4	DOFA Large	.474	.647
5	OlmoEarth Base	.469	.647

Efficiency — Throughput vs Accuracy

PR #60-#80

GPU throughput (img/s) — m-eurosat

Model	img/s	GFLOPs	Acc
ResNet-50 MoCo	3 193	8	.76
DOFA Base	1 747	36	.71
Terramind Base	1 680	36	.68
OlmoEarth Tiny	780	9	.73
OlmoEarth Nano	789	2	.67
DOFA Large	581	124	.73
DINOv3-SAT ViT-L	346	165	.74

Pareto surprises:

- WINNER

ResNet-50 MoCo — 3 193 img/s, 8 GFLOPs, 23M params, yet **.76 avg acc.** Matches or beats every ViT-scale model while running 5–10× faster.
- EFFICIENT

OlmoEarth Nano — 3.6M params, 0.6 GB peak VRAM, 1.6 GFLOPs. Matches Panopticon accuracy at 5× lower cost.
- EFFICIENT

OlmoEarth Tiny (14M) ties DOFA Large (337M) and DINOv3-SAT (304M) at 2× the throughput.
- WORST VALUE

Prithvi v1/v2 — 1 700 img/s, 35 GFLOPs, only .56–.60 acc. Same throughput tier as DOFA Base but far lower accuracy.

Intrinsic Dimensionality Analysis

METHOD: INTRINSIC_DIM

id_TwoNN avg · norm. acc = rank-normalized within each dataset

Model	id_TwoNN	norm. acc
EarthLoc ResNet-50	44	27%
OlmoEarth Large	22	85%
Prithvi v2 100M CLS	18	53%
DOFA Large	17	85%
DINOv3-SAT ViT-L	16	77%
DOFA Base	15	78%
OlmoEarth Nano (128-d)	14	75%

Pattern: id > 14 → consistently 72–85% norm. acc. Higher intrinsic dim correlates with better downstream accuracy.

TASK GAP **EarthLoc ResNet-50** — id = 44 but only 27% norm. acc. Geo-localization training, not classification — high dimensionality without discriminative structure.

SURPRISE **OlmoEarth Nano** — 128-d output, id_TwoNN = 14, matching DINOv3-SAT ViT-L (1024-d). Same intrinsic complexity in 8× fewer dimensions.

FLAT **Prithvi cluster** — id ≈ 7, geometrically flat despite 768–1280-d features. Explains consistently low accuracy.

Calibration Metrics

PR #105

ECE / RMS-CE / MCE on every classification probe, plus a temperature-scaling baseline (Guo et al., 2017):

```
1  eval:
2    calibration:
3      n_bins_knn: null # null → knn_k + 1
4      n_bins_linear: 15
5      temp_scale: true # +ece_ts / mce_ts
6                      # + temperature
```

```
1  # fit T on val logits, NLL via LBFGS
2  T = fit_temperature(val_logits, y_val)
3  # T > 1 overconfident · T < 1 under
```

KNN probs quantize to $k+1$ levels → binned at knn_k+1 . TS applies to Linear only.

Linear probe — m-eurosat (RGB)

Model	ECE	ECE·TS	T
DOFA Large	.007	.021	0.23
DINOv3-SAT ViT-L	.016	.030	0.18
ResNet-50 MoCo	.018	.032	0.34
OlmoEarth Base	.032	.054	0.71
DOFA Base	.043	.031	0.21

FINDING

Probes are already well-calibrated ($ECE < .05$). Fitted $T < 1$ → logits *underconfident*, so temperature scaling often **raises** ECE. DOFA Base is the lone beneficiary.

Key Findings

🌐 **S2 multispectral** (EuroSAT, BigEarthNet, So2Sat)

→ **OlmoEarth** dominates KNN + Linear

🌲 **RGB / aerial** (ForestNet, PV4GER, Brick Kiln)

→ **DINOv3-SAT** or **Panopticon** lead

🌿 **Multi-label** (BigEarthNet, BENV2, TreeSatAI)

→ OlmoEarth top KNN; Terramind competitive linear

⚡ Terramind **Base** beats Terramind **Large** on So2Sat linear (0.739 vs 0.712)

⚡ CLS token **collapses** on EuroSAT for Prithvi/Clay — patch-mean wins

⚡ **ImageStats** (raw pixel stats) still hits .95 on m-brick-kiln — low semantic difficulty, though GeoFMs now edge it out

⚡ **SatlasPretrain Swin** (RGB-only) peaks at m-pv4ger #9 / m-brick-kiln #16 — RGB backbones trail multispectral FMs on S2

⚡ **OlmoEarth v1.1** matches v1 at $\approx 3\times$ fewer MACs; Tiny/Nano gain most on So2Sat + BigEarthNet

⚡ Linear probes are **well-calibrated out of the box** (ECE < .05) — temperature scaling rarely helps